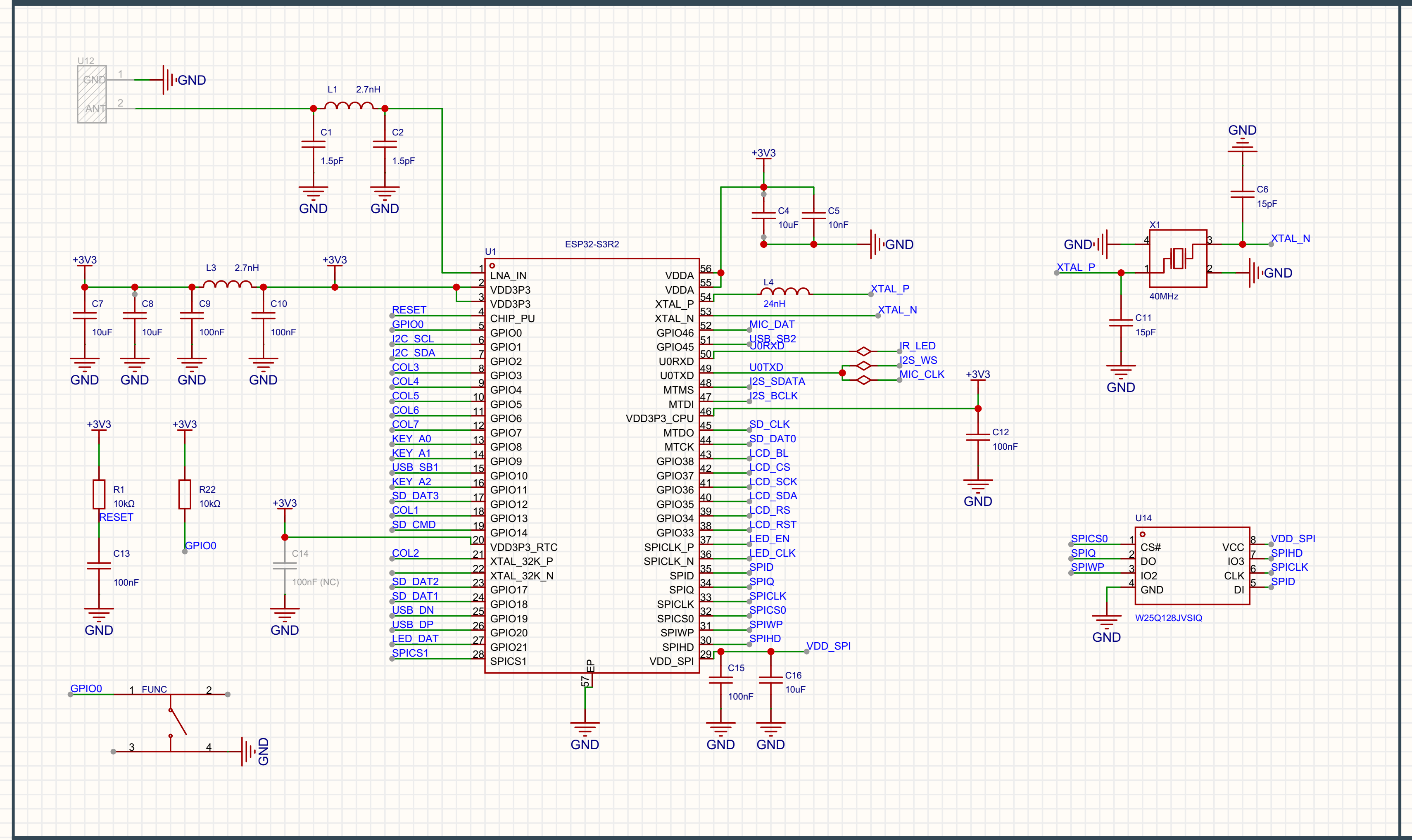
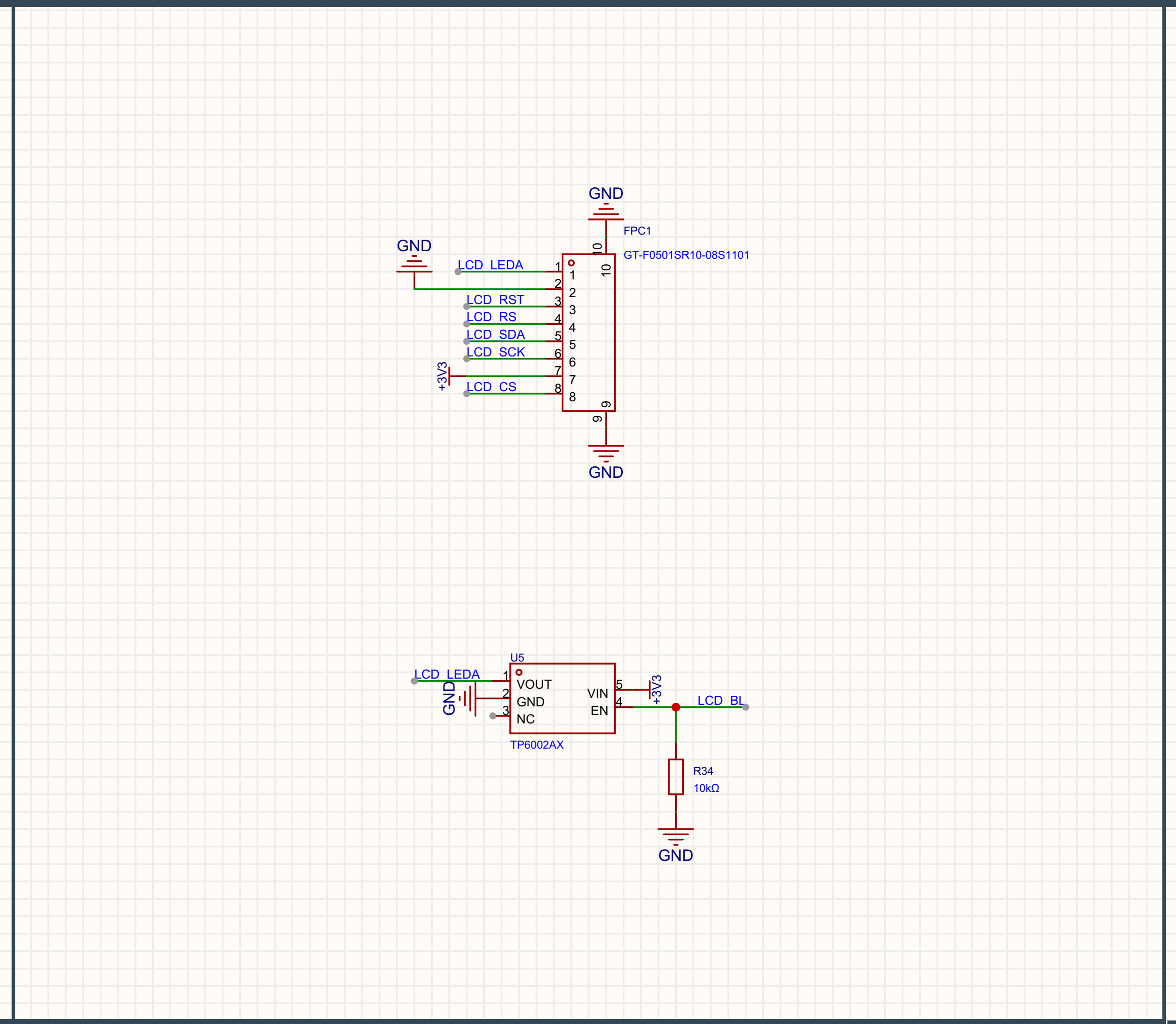
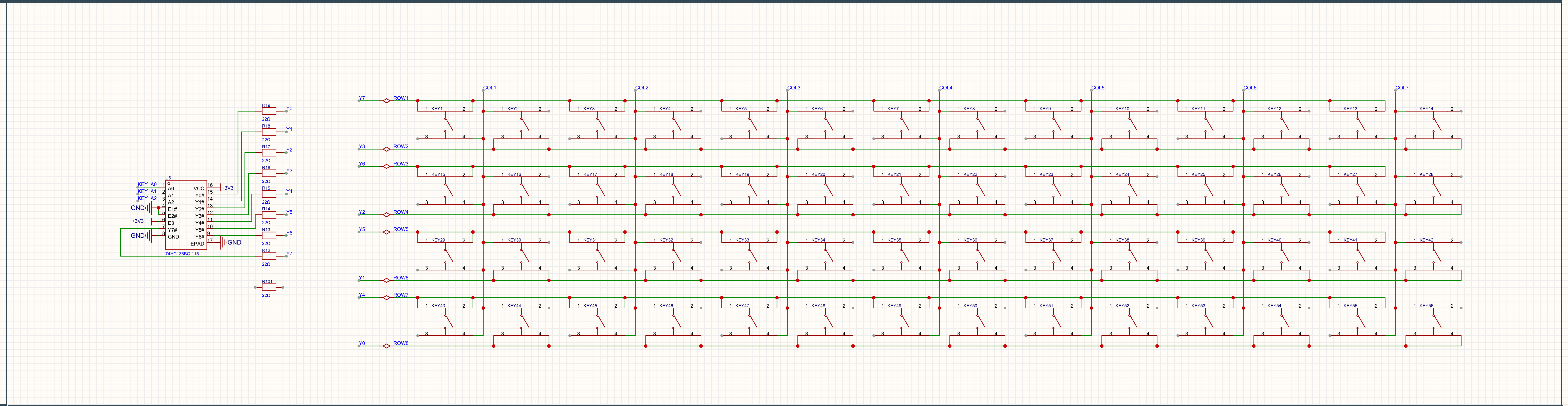
01 PROCESSOR / ESP32-S3


ESP32-S3R2
GND
+3V3
40MHz
XTAL_P
XTAL_N
W25Q128JVSIQ

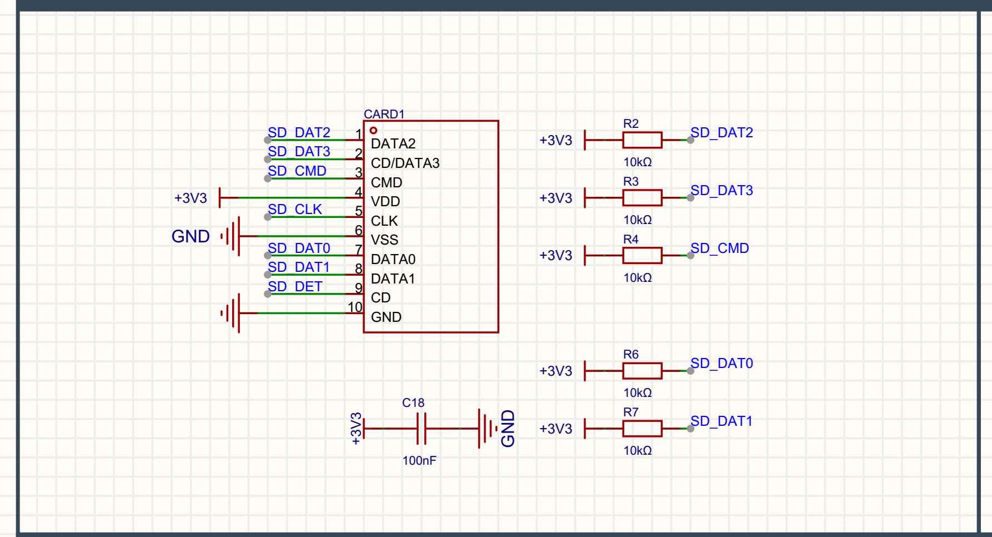
02 TFT DISPLAY / BACKLIGHT


GND
LCD_BL
TPS61169

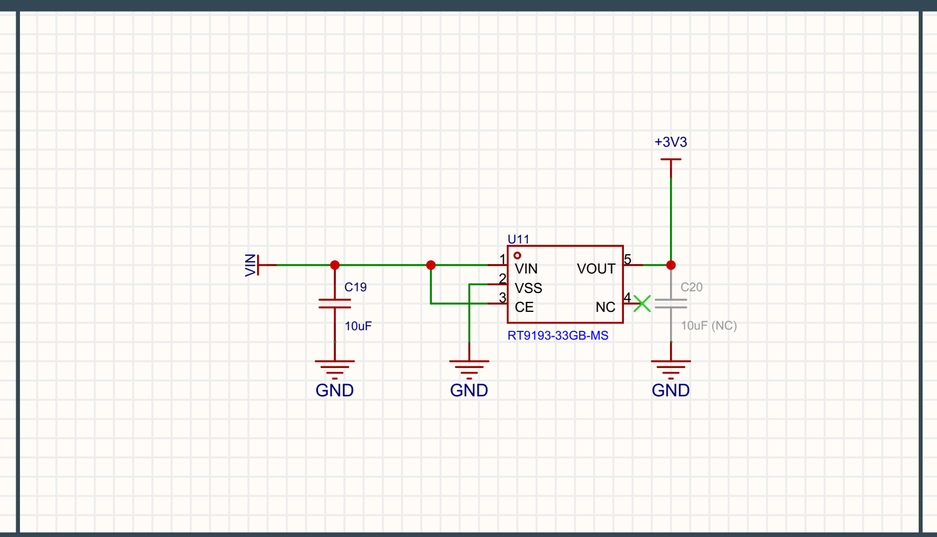
03 KEY MATRIX / 56 KEYS


COL1
COL2
COL3
COL4
COL5
COL6
COL7
ROW1
ROW8
74HC138BQ,115

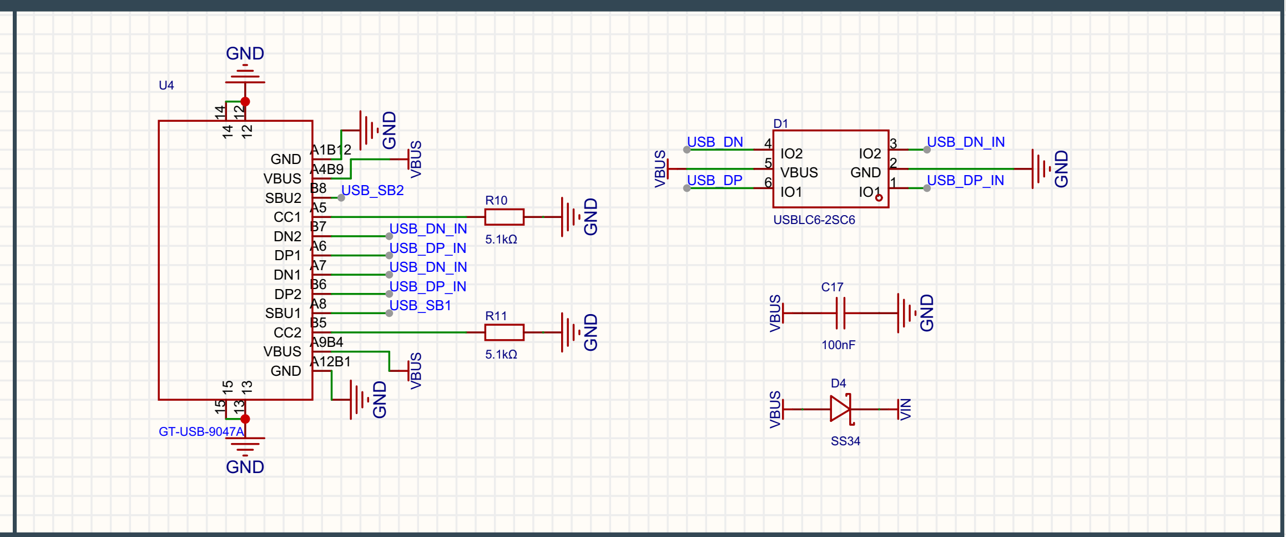
04 microSD STORAGE



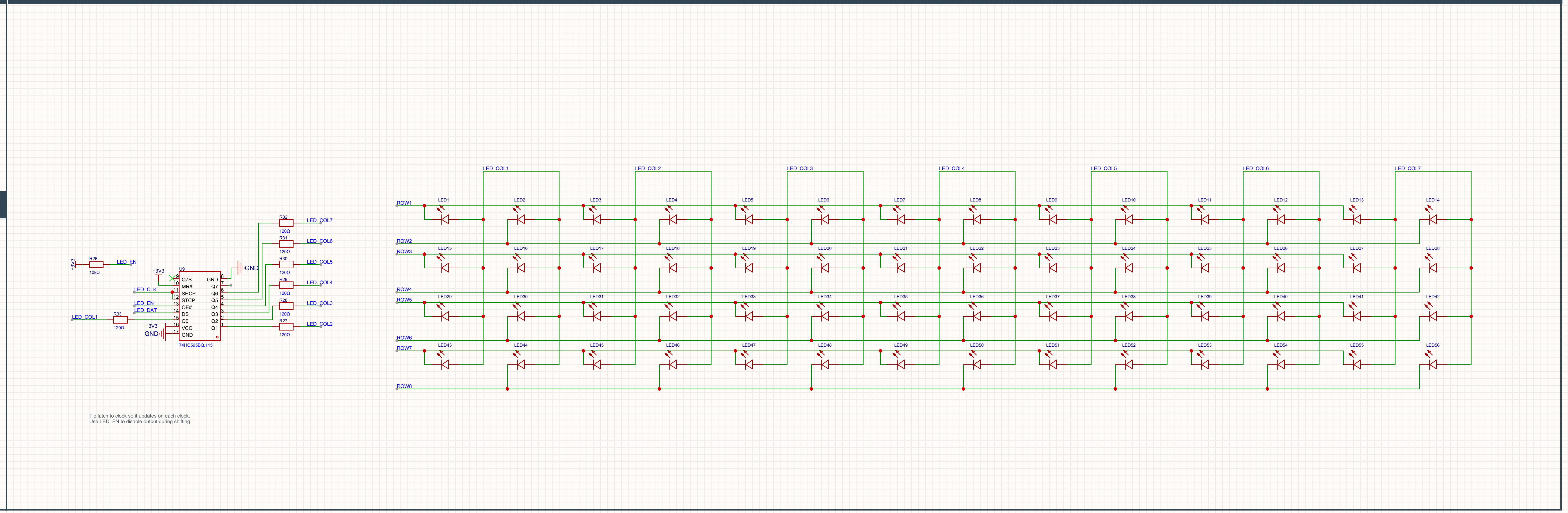
05 3.3 V POWER



06 USB-C / ESD

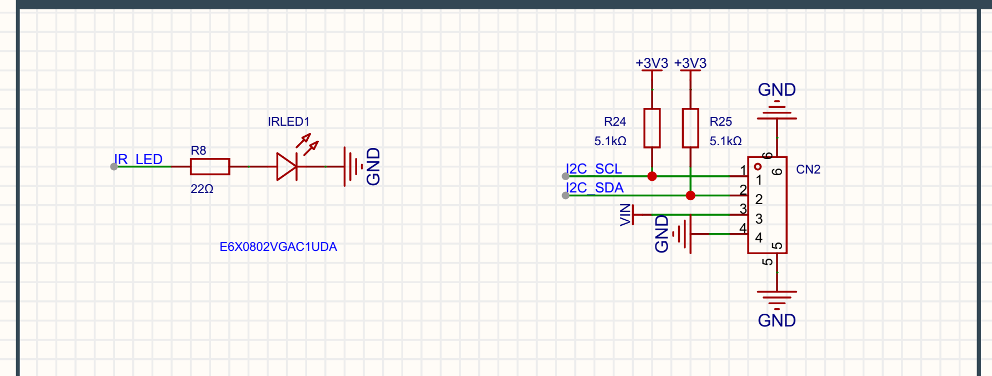


07 LED MATRIX / 56 LEDs

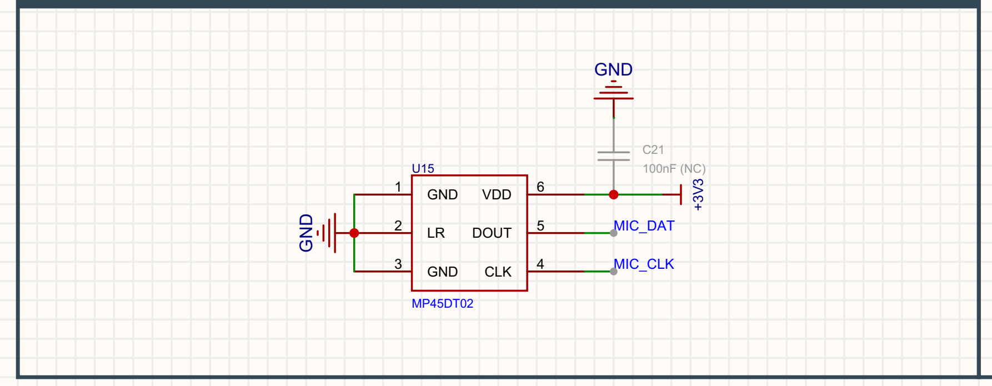

LED_COL1
LED_COL2
LED_COL3
LED_COL4
LED_COL5
LED_COL6
LED_COL7
74HC595BQ,115

Tie latch to clock so it updates on each clock.
Use LED_EN to disable output during shifting

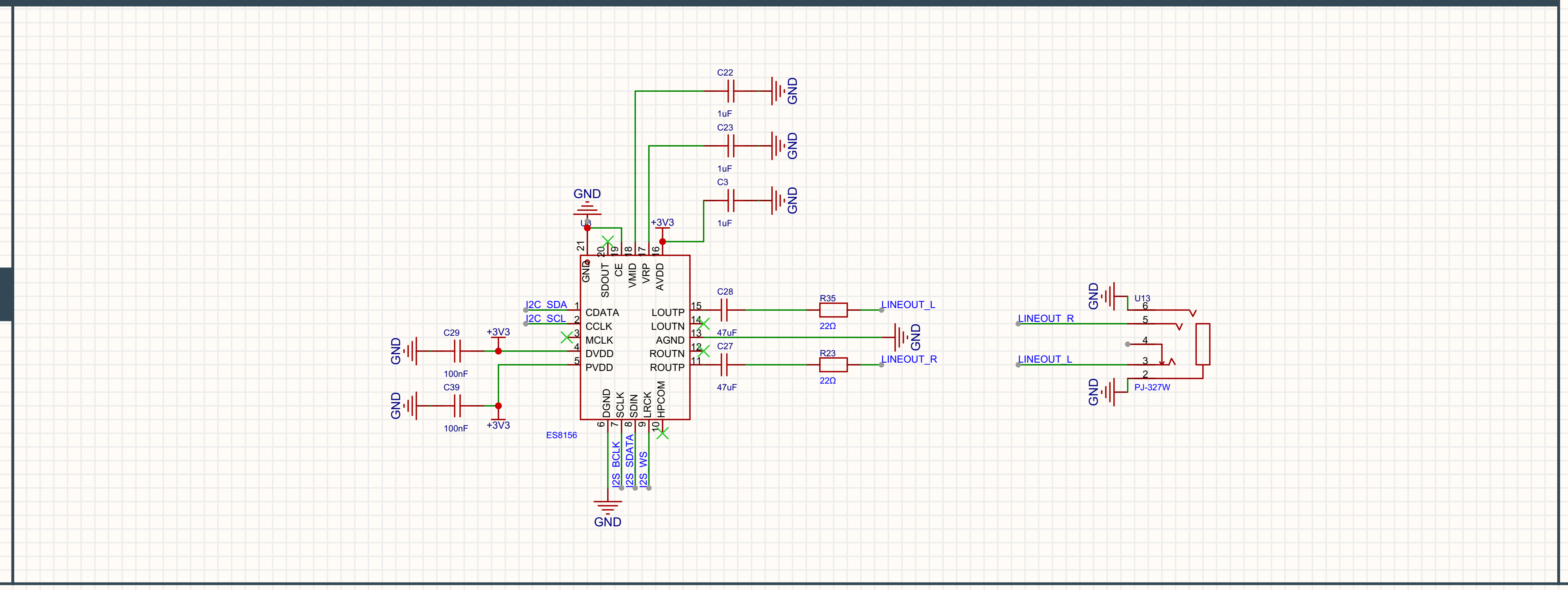
08 AUX / I2C + IR



09 DIGITAL MICROPHONE



10 AUDIO DAC / STEREO OUT


ES8156
LINEOUT_L
LINEOUT_R
PJ-327M

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SYSTEMS	V1.0	A4	203 Systems